

Evaporative Emissions



The Evaporative Emission (EVAP) system consists of the following components:

- EVAP canister (located in front of the fuel tank)
- EVAP canister purge valve (located on the intake manifold)
- Dual check valve (3.5L Gasoline Turbocharged Direct Injection (GTDI) engine) (part of the canister purge valve)
- Vapor ejector (3.5L GTDI engine) (located in front of the engine)
- EVAP canister vent solenoid and dust separator assembly (located on the front of the EVAP canister)
- Fuel Tank Pressure (FTP) sensor and vapor tube assembly (located on top of the fuel tank)
- Vapor tubes
- Fresh air hose (located on top of the transmission)

The EVAP system:

- prevents hydrocarbon emissions from reaching the atmosphere.
- stores fuel vapors in the EVAP canister that are generated during vehicle operation or hot soak, until they can be consumed by the engine during normal engine operation.
- routes the stored fuel vapors to the engine during engine operation.
- is controlled by the PCM which, using various sensor inputs, calculates the desired amount of purge flow. The PCM regulates the purge flow, induced by the application of intake manifold vacuum, by varying the duty cycle applied to the EVAP canister purge valve.

The fuel vapors are routed:

- from the fuel tank through the fill limit valve and fuel vapor vent valve. The fuel vapor vent valves are part of the fuel tank assembly and cannot be serviced separately.
- to the EVAP canisters through a vapor line.
- to the engine when the EVAP canister purge valve is opened by the PCM.

The EVAP canister:

- contains activated carbon.
- stores fuel vapors.

The EVAP canister purge valve:

- is normally closed.
- regulates the purging of the EVAP canister.
- is controlled by the PCM.

The dual check valve (3.5L GTDI engine):

- routes fuel vapors to the intake manifold when the intake manifold is not pressurized.
- routes fuel vapors to the vapor ejector when the intake manifold is pressurized.

The vapor ejector (3.5L GTDI engine):

- is a venturi valve.
- uses pressurized air from the RH turbocharger-to-Charge Air Cooler (CAC) tube to route fuel vapors to the Air Cleaner (ACL) outlet tube-to-LH turbocharger tube.

The EVAP canister vent solenoid:

- is controlled by the PCM.
- is normally open.
- seals the EVAP system for the inspection and maintenance (I/M 240) test and OBD II leak and pressure tests.

The EVAP dust separator:

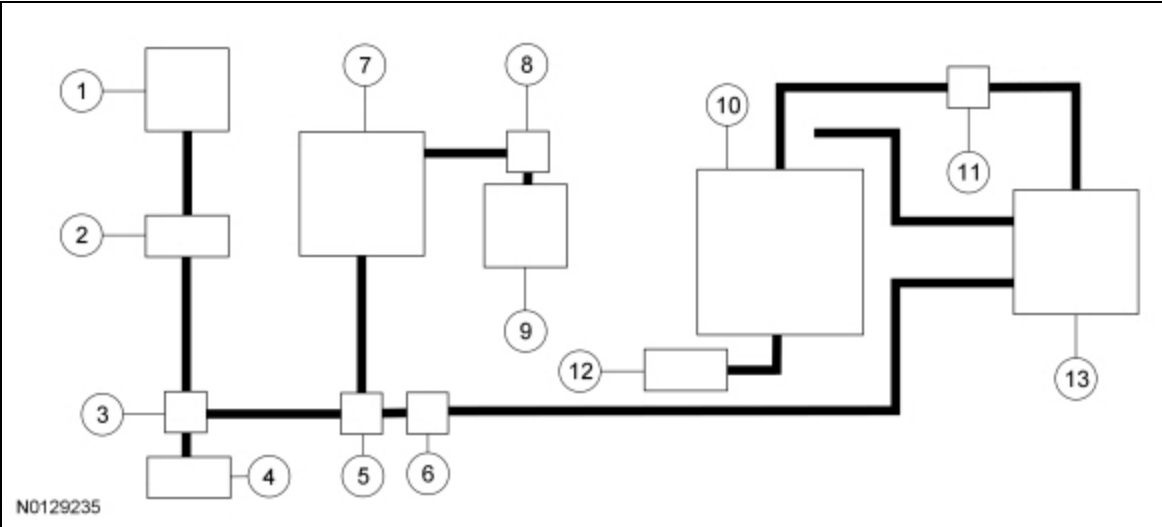
- prevents suspended dust and dirt particles from entering the EVAP system.
- is serviced as a separate item.

The FTP sensor:

- measures the pressure levels in the fuel tank continuously.
- serves as an input to the PCM.

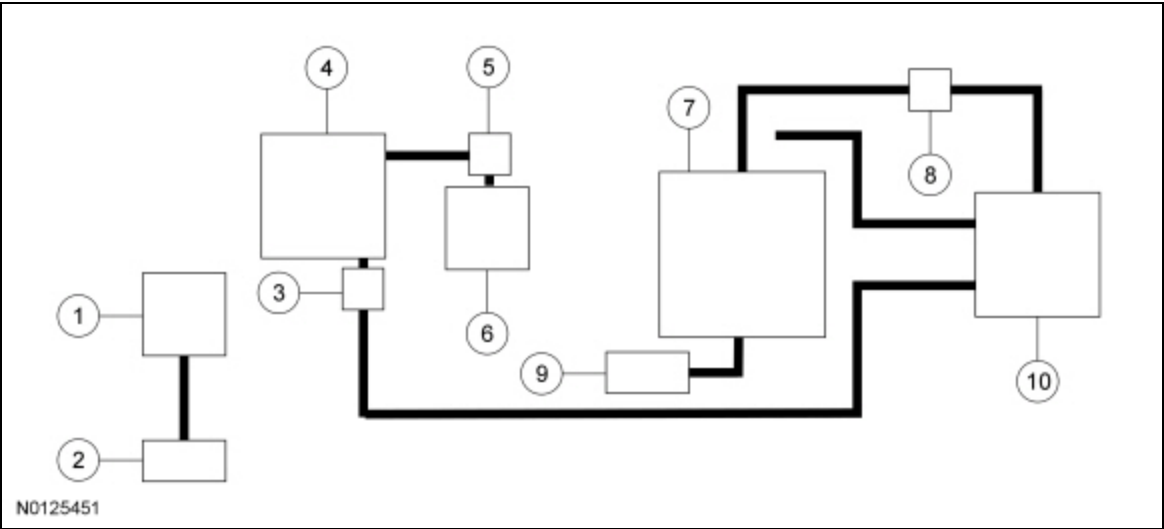
Vehicle Emission Vacuum Routing

3.5L Gasoline Turbocharged Direct Injection (GTDI)



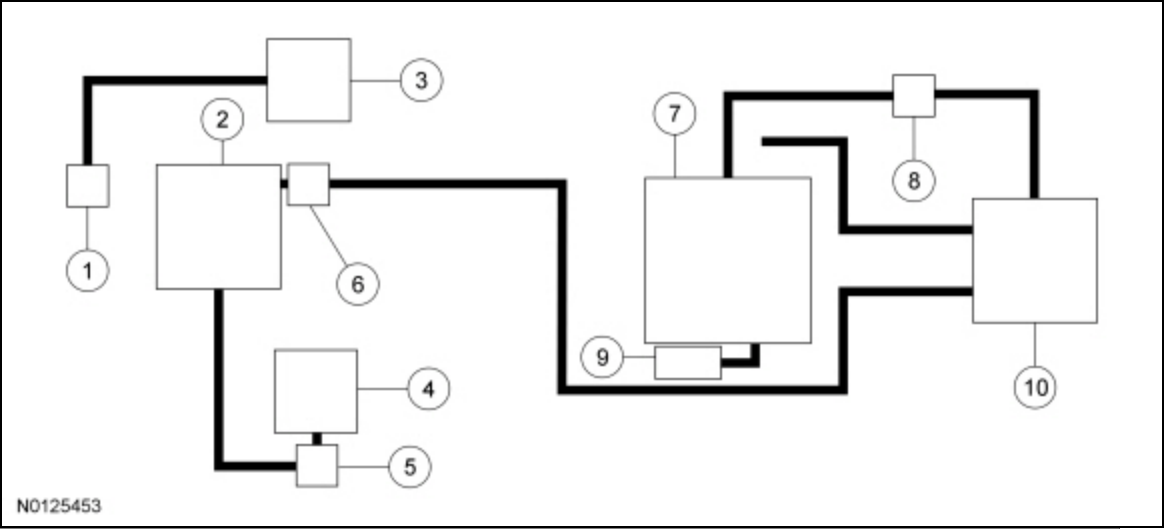
Item	Part Number	Description
1	6A5056A505	LH valve cover
2	—	Air Cleaner (ACL) outlet tube-to-LH turbocharger tube (part of 9C623)
3	—	Vapor ejector (part of 9C915)
4	9C6469C646	RH turbocharger-to-Charge Air Cooler (CAC) tube
5	—	Dual check valve (part of 9B325)
6	9B3259B325	Evaporative Emission (EVAP) canister purge valve
7	94249424	Intake manifold
8	6A6666A666	PCV valve
9	65826582	RH valve cover
10	90029002	Fuel tank
11	—	Fuel Tank Pressure (FTP) sensor (part of 9E325)
12	90349034	Fuel tank filler pipe
13	9E8579E857	EVAP canister

3.7L



Item	Part Number	Description
1	6A5056A505	LH valve cover
2	9R5309R530	Air Cleaner (ACL) outlet tube
3	9G8669G866	Evaporative Emission (EVAP) canister purge valve
4	94249424	Intake manifold
5	6A6666A666	PCV valve
6	65826582	RH valve cover
7	90029002	Fuel tank
8	—	Fuel Tank Pressure (FTP) sensor (part of 9E325)
9	90349034	Fuel tank filler pipe
10	9E8579E857	EVAP canister

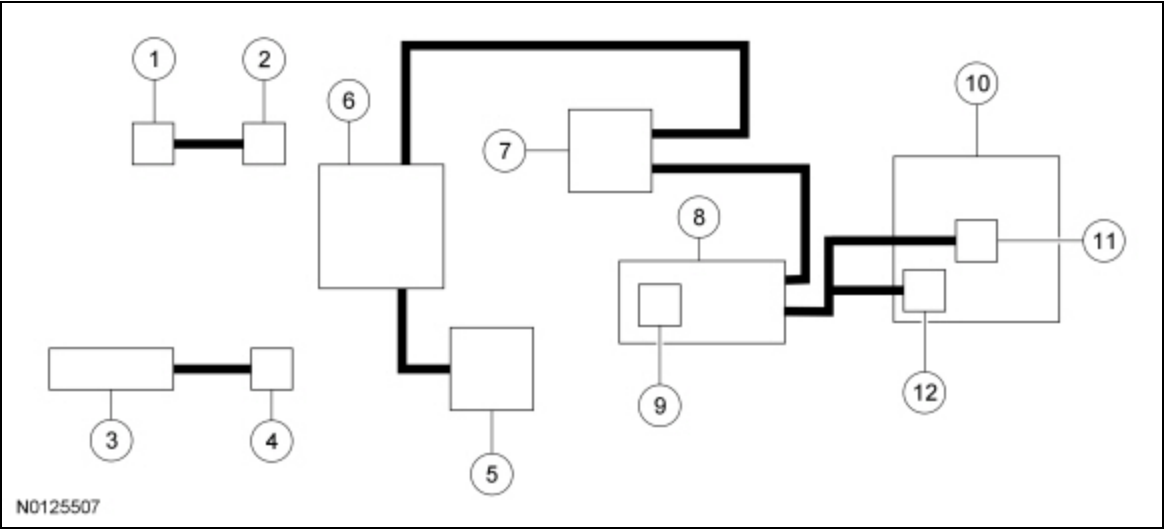
5.0L



Item	Part Number	Description
1	9B6599B659	Air Cleaner (ACL) outlet pipe

2	9425 9425	Intake manifold
3	6582 6582	RH valve cover
4	6A505 6A505	LH valve cover
5	—	PCV valve (part of LH valve cover)
6	9G866 9G866	Evaporative Emission (EVAP) canister purge valve
7	9002 9002	Fuel tank
8	—	Fuel Tank Pressure (FTP) sensor (part of 9E325)
9	9034 9034	Fuel tank filler pipe
10	9E857 9E857	EVAP canister

6.2L



Item	Part Number	Description
1	6A666 6A666	PCV valve
2	9F624 9F624	PCV fitting
3	6A505 6A505	LH valve cover
4	9A589 9A589	Air Cleaner (ACL) outlet pipe-to-Throttle Body (TB) adapter
5	—	Brake booster
6	9S455 9S455	Intake manifold
7	9C915 9C915	Evaporative Emission (EVAP) canister purge valve
8	9D653 9D653	EVAP canister assembly
9	9F675 9F675	EVAP canister vent solenoid and dust separator assembly
10	9K007 9K007	Fuel tank
11	—	Fuel pump module
12	—	Fuel vent valve